MCO-1D — BILL OF MATERIALS v13 ADDENDUM

New hardware items since BOM v12

This addendum adds three new items required for per-wing AoA closed-loop control and stall detection. These items are required before any flight testing of hypotheses H-24, H-25, H-26 (per-wing AoA differential configurations) and H-27 (servo bandwidth mapping).

NEW BOM ITEMS — PER-WING AoA FEEDBACK SYSTEM

1	pcs	MCP3208 8-channel 12-bit SPI ADC	Microchip MCP3208-CI/P or equivalent. Reads all 8 pitch servo feedback potentiometers on a single SPI bus. SPI Bus 1, CE1. Speed: 1MHz. VREF: 3.3V from Pi. Resolution: 12-bit over 3.3V → 0.8mV/step → ~0.044°/step at servo. Sample rate: reads all 8 channels in ~107µs — negligible at 50Hz. Channel assignment: CH0=W1p, CH1=W1s, CH2=W2p, CH3=W2s, CH4=W3p, CH5=W3s, CH6=W4p, CH7=W4s. Enable via: dtoverlay=spi1-1cs in /boot/config.txt.	~$3
8	wires	Servo feedback signal wires	One wire per pitch servo feedback pin to MCP3208 ADC channel. Keep short (<30cm) to reduce noise pickup near motor wiring. Use twisted pair with ground return if longer runs needed. The feedback potentiometer output is 0.5-2.5V over servo travel. No pull-up or pull-down resistors needed — ADC input impedance is adequate for the servo potentiometer source impedance.	<$1
1	pcs	ADS1115 16-bit I2C ADC (optional, alternative)	Alternative to MCP3208 if SPI Bus 1 is unavailable. Need 2 chips (4 channels each) to cover 8 wings. Addresses: 0x48 and 0x49 (ADDR pin tied to GND and VCC). 16-bit resolution over 4.096V PGA range. Maximum rate: 860 SPS per chip — adequate at 50Hz control loop. MCP3208 is recommended (single chip, faster, cleaner integration).	~$2 each

Wiring Diagram (text description)

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Pi SPI1 CE1 (GPIO7) → MCP3208 /CS
Pi SPI1 MOSI (GPIO10) → MCP3208 DIN
Pi SPI1 MISO (GPIO9) → MCP3208 DOUT
Pi SPI1 SCLK (GPIO11) → MCP3208 CLK
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Pi 3.3V → MCP3208 VDD and VREF
Pi GND → MCP3208 GND
W1p servo feedback pin → MCP3208 CH0 (via 1kΩ series resistor)
W1s servo feedback pin → MCP3208 CH1
... W4s servo feedback pin → MCP3208 CH7
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✓ NOTE: Add a 1kΩ series resistor on each feedback signal line as protection against inadvertent shorts to the 5V servo rail. The series resistor does not significantly affect accuracy given the ADC input impedance.

Updated Software Statistics	
Wave Library lines	43,318
Active hypotheses	24 (H-01 to H-27 excluding H-09/10/11)
H-27 variants	12 (servo bandwidth at 4 frequencies × 3 airspeeds)
New GS commands	servo_health_report, lift_balance_report, aoa_cl_enable, aoa_cl_disable, calibrate_servo_feedback